

ECON345 Homework 6: Hypothesis Testing & Correlation

Due: October 8th, 11:59 PM

Develop your code in an R script first if that helps you think, then answer the questions (including your code) in an R Markdown file. Comment your code to explain what you're doing – partial credit rewards a reasonable, well-documented attempt, not just a correct final answer. For any part you can't produce output for, briefly explain what you tried and what went wrong.

Data and Codebook

These questions use data from the World Bank's 2014 Enterprise Survey of firms in India, in `india_enterprise_survey_2014_clean.csv`.

Three variables share the same response scale, describing how much of an obstacle a given factor poses to the firm's current operations: `obstacle_electricity`, `obstacle_political_instability`, and `obstacle_access_to_finance`. Each is coded 0 (no obstacle), 1 (minor obstacle), 2 (moderate obstacle), 3 (major obstacle), or 4 (very severe obstacle).

Three more variables round out the dataset:

- `top_manager_female` – is the top manager female? (0 = No, 1 = Yes)
- `first_year_operations` – the first year the firm operated (ranges 1863–2013)
- `percent_of_sales_direct_exports` – what percent of total sales are direct exports, i.e. sales where the immediate recipient is outside the country (ranges 0–100)

Question 1 (10 points)

A professor studying firms in India wants to compare two groups: recently established firms (started operating in 1990 or later) and older firms (started before 1990). She's interested in whether the two groups perceive political instability as an obstacle differently.

Question 1.1

Create a new variable `recent_firm` that equals 1 if a firm's first year of operation is 1990 or later, and 0 otherwise.

Question 1.2

Which test will you use, and why? (Options: one-sample two-sided t-test, one-sample one-sided t-test, two-sample two-sided t-test, two-sample one-sided t-test.)

Question 1.3

State the null and alternative hypotheses.

Question 1.4

Suppose $\alpha = 0.05$. Run the test and verify that your code produces the output below (if you can't reproduce it, you can still answer 1.5):

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
    <dbl> <dbl>   <dbl>   <chr>         <dbl>   <dbl>   <dbl>
1    -4.57 4256. 0.00000508 two.sided    -0.114 -0.163 -0.0651
```

Question 1.5

Interpret the results (reference the p -value and α). What factors might be contributing to this result?

Question 2 (10 points)

The same professor speculates that women-led firms (a woman as top manager) are more worried about access to finance as an obstacle than firms led by men.

Question 2.1

Which test will you use, and why?

Question 2.2

State the null and alternative hypotheses.

Question 2.3

Suppose $\alpha = 0.05$. Run the test and verify your code produces this output (if not, you can still answer 2.4 and 2.5):

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
    <dbl> <dbl>   <dbl>   <chr>         <dbl>   <dbl>   <dbl>
1    -2.28  851. 0.0114 less        -0.0864 -Inf    -0.0240
```

Question 2.4

Interpret the results — what do they mean for the professor's question?

Question 2.5

What factors might be contributing to this result?

Question 3 (10 points)

A friend claims that 5.8% of firms' sales, on average, are direct exports. You want to test whether that's true, or whether the true percentage is something other than 5.8%. (Values are reported from 0 to 100.)

Question 3.1

Which test will you use, and why?

Question 3.2

State the null and alternative hypotheses.

Question 3.3

Suppose $\alpha = 0.05$. Run the test and verify your code produces this output (if not, you can still answer 3.4 and 3.5):

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
    <dbl> <dbl>   <dbl>   <chr>         <dbl>   <dbl>   <dbl>
1      1.80  9280  0.0715 two.sided      6.18    5.77    6.59
```

Question 3.4

Interpret the results (reference the p -value and α).

Question 3.5

What factors might be contributing to this result?

Question 4 (10 points)

A businessman writes an opinion piece claiming India has a serious electricity problem – specifically, that lack of electricity ranks as a “major obstacle,” a 3.0 on our 0–4 scale. We're skeptical, and want to test whether firms are, on average, *less* worried than he claims.

Question 4.1

Which test will you use, and why?

Question 4.2

State the null and alternative hypotheses.

Question 4.3

Suppose $\alpha = 0.05$. Run the test and verify your code produces this output (if not, you can still answer 4.4 and 4.5):

```
# A tibble: 1 × 7
  statistic t_df p_value alternative estimate lower_ci upper_ci
  <dbl> <dbl> <dbl> <chr> <dbl> <dbl> <dbl>
1 -64.9 9275 0 less 2.26 -Inf 2.28
```

Question 4.4

Interpret the results (reference the p -value and α).

Question 4.5

What factors might be contributing to this result?

Question 5 (35 points)

For this question, consider the relationship between `obstacle_access_to_finance` and `obstacle_political_instability`.

Question 5.1

Prior to computing the correlation coefficient, state what you believe the relationship between the two variables to be (talk about both strength and sign). Provide reasoning behind your statement. (10)

Question 5.2

Report the correlation coefficient. If you're using `cor()`, be sure to include `use = "complete.obs"` to account for NAs. Verify that your code produces the output below: (10)

```
# A tibble: 1 × 1
  cor
  <dbl>
1 0.197
```

Question 5.3

Is this correlation coefficient consistent with what you expected in 5.1? (5)

Question 5.4

Create a scatterplot to visualize the relationship between the two variables. Because both variables are discrete, a standard scatterplot is not very informative — consider a geometry that conveys how many observations share a point (`geom_count()`), or one that jitters points to reduce overplotting (`geom_jitter()`). (10)

Submission

Upload two files to Canvas before the deadline:

- your `week_06.Rmd`, and

- the **knitted PDF**.

As always: comment your code, and if you get stuck, show your work. If your `.Rmd` won't knit, get each piece working before you move on to the next, and restart R and knit clean if results look stale.